T7.2.3-Worksheet 1-Flow Charts

Answer the following questions.

1. Draw a flow that
   - Inputs a number
   - Find out number is even or odd (using MOD function)
   - Output Even or Odd

2. A programmer is designing a program to check the length of a password and to check password input is the same as the stored password.

   The program requirements are:
   - input the password, `Password`
   - check if there are at least 8 characters in the password
   - check that the password is **not** the same as the stored password `OldPass`
   - output 'accepted' if both tests are completed successfully
   - otherwise, output 'rejected'.

   Use the variable names given.

3. An algorithm needs to total 50 numbers between 1 and 100 inclusive.

   Draw a flowchart that:

   - Uses a count-controlled loop from 1 to 50
   - Uses an appropriate prompt to ask for a number between 1 and 100
   - Totals the numbers as they are entered
   - Outputs the total after the loop has completed with an appropriate message.

T7.2.3-Worksheet 1-Flow charts -- Answers

Answer the following questions.

1. Draw a flow that
   - Inputs a number
   - Find out number is even or odd (using MOD function)
   - Output Even or Odd

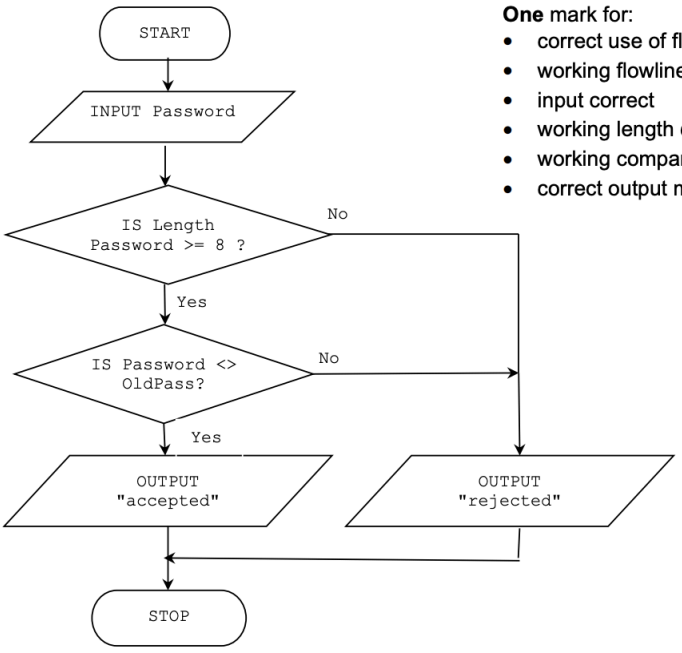
Answer based on the discussion in class -- see last page

2. A programmer is designing a program to check the length of a password and to check password input is the same as the stored password.

   The program requirements are:
   - input the password, `Password`
   - check if there are at least 8 characters in the password
   - check that the password is **not** the same as the stored password `OldPass`
   - output 'accepted' if both tests are completed successfully
   - otherwise, output 'rejected'.

   Use the variable names given.

Answer:



One mark for:
- correct use of flowchart symbols
- working flowlines
- input correct
- working length check
- working comparison
- correct output messages

3. An algorithm needs to total 50 numbers between 1 and 100 inclusive.

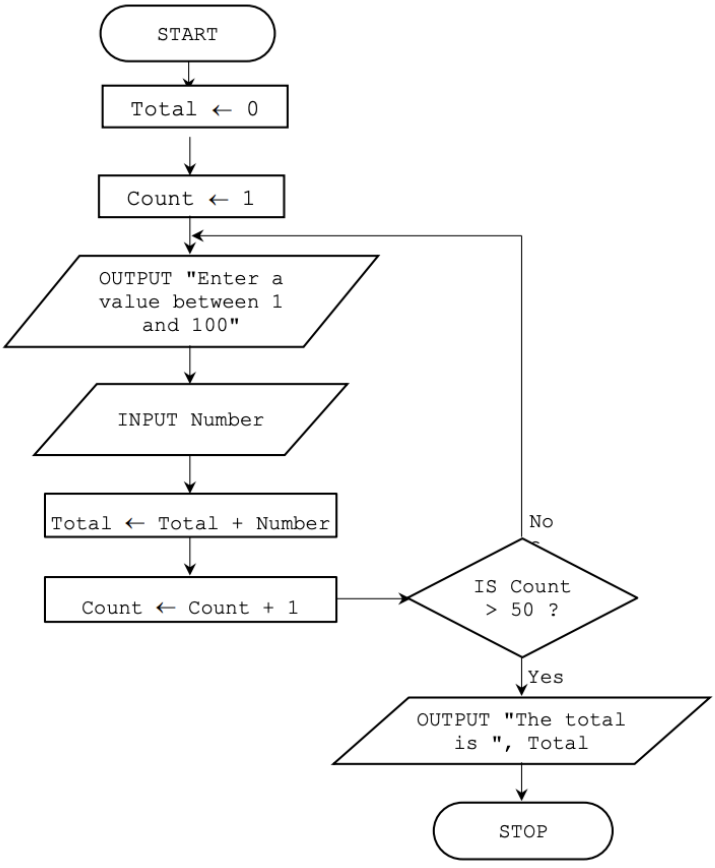
Draw a flowchart that:

- Uses a count-controlled loop from 1 to 50
- Uses an appropriate prompt to ask for a number between 1 and 100
- Totals the numbers as they are entered
- Outputs the total after the loop has completed with an appropriate message.

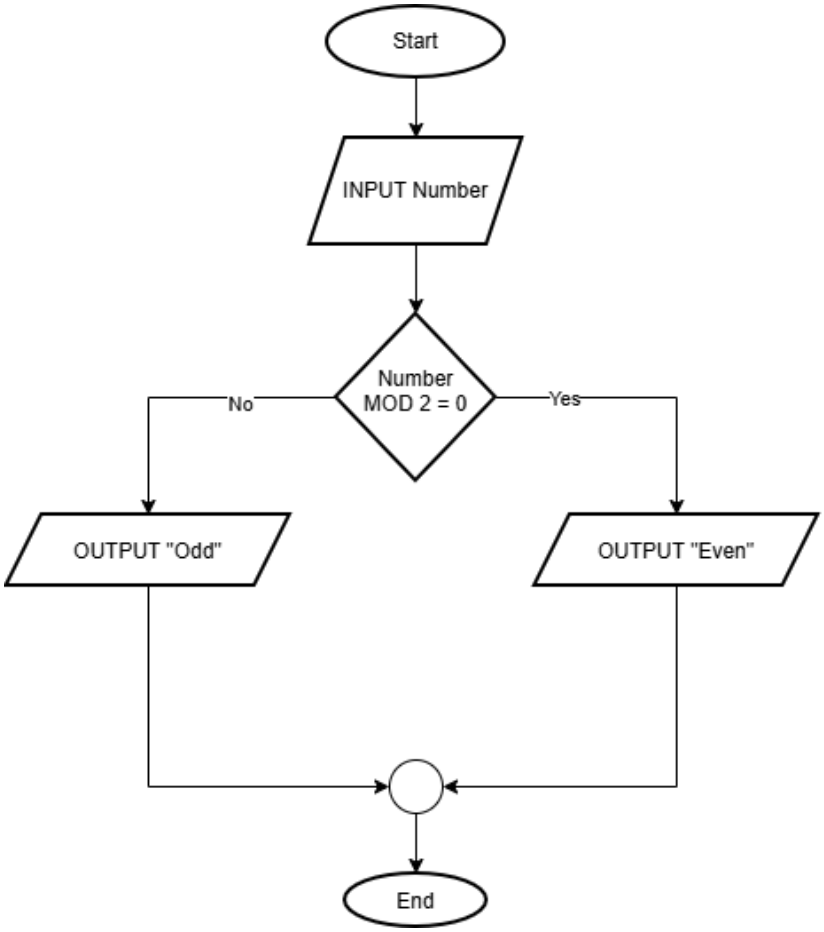
Answer:

One mark per mark point

- Two labelled terminators START and STOP
- Loop initialisation, incrementation and termination
- Input with prompt
- `Total` initialised and totalling of inputs
- Output with message
- Fully correct flowchart with all correct arrows, yes/no labels and symbols



1. Answer:



This flow chart determines and outputs whether the input number is even or odd.